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Tendons and Ligaments: Connecting Developmental Biology to Musculoskeletal Disease Pathogenesis

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Abstract

Tendons and ligaments provide connections between muscle and bone or bone and bone to enable locomotion. Damage to tendons and ligaments caused by acute or chronic injury or associated with aging and arthritis is a prevalent cause of disability. Improvements in approaches for the treatment of these conditions depend on a better understanding of tendon and ligament development, cell biology and pathophysiology. This review focuses on recent advances in the discovery of transcription factors that control ligament and tendon cell differentiation, how cell and extracellular matrix homeostasis are altered in disease and how this new insight can lead to novel therapeutic approaches.

Keywords

tendon; ligament; arthritis; Mxk; Scx; Egr1/2

INTRODUCTION

Despite the unique and important roles of tendons and ligaments in the musculoskeletal function and musculoskeletal diseases, research in this area is not as advanced as in other skeletal tissues. This is partly because, until recently, the transcription factors critical for the formation and maintenance of these tissues were unknown. The recent identification of Scleraxis (Scx), Mohawk (Mxk) and early growth response factor (Egr1) and their specific roles in tendon and ligament development, maturation, homeostasis and repair of mature

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tissues has substantially advanced the field of tendon and ligament research. This review summarizes the current knowledge and concepts regarding the development, function and disease-associated changes in tendon and ligament cells and extracellular matrix (ECM) with a focus on Mx, Scx and Egr1.

Tendons and ligaments are distinct but closely related tissues. Although developmental processes of tendon and ligament tissues and the phenotypes of mice with genetic deletion of the key transcription factors are different (see below section 2 and 3), the basic structures of these tissues and the gene expression profiles of their major cell type (tenocytes) are rather similar.

We therefore focus this report on the common basis between tendons and ligaments uncovered in recent research and use anterior cruciate ligament (ACL) injuries as a representative example of tendon and ligament injuries (see section 1 below).

1. Clinical significance of tendon and ligament tissue damage

Tendon and ligament damage caused by injury or overuse or associated with aging and arthritis is a common clinical problem because damaged tissues heal very slowly and rarely recover completely (1). Damage to the glenohumeral ligaments and the rotator cuff tendons of the shoulder joint, which can lead to shoulder pain and functional limitations, are highly prevalent, with >50% of individuals older than 60 years of age being affected (2). The success of surgical treatment is limited, as up to 50% of rotator cuff repair procedures fail.

The Achilles tendon is the most frequently ruptured tendon in humans (3). Injury and aging-related changes in cells and the ECM of tendon and ligament in the knee joints, particularly the ACL, represent major risk factors for the development of osteoarthritis (OA) (4, 5). The ACL is essential for knee kinematics, especially in rotation, and functions as an anterior/posterior stabilizer (6, 7). In the setting of ACL deficiency, the articular cartilage as well as the menisci are more susceptible to arthritic changes. A large number of OA patients without a history of ligament injury have ACL deficiency at the time of total knee arthroplasty (8), and a correlation between the radiologic OA grade and the histological grade of ACL degeneration has been reported in end-stage OA (9). In addition, ACL rupture is more common among patients with symptomatic knee OA than in those without. It has been reported that fewer than half of subjects with ACL rupture recall a knee injury, suggesting that this risk factor for knee OA is under-recognized (8). These findings indicate that substantial changes occur in tendons and ligaments in arthritic knees that are not related to major trauma but are presumably caused by mediators, such as cytokines or matrix degrading enzymes, in the arthritic joints that affect tendon and ligament cells and the ECM.

2. Biology of mature tendon and ligament tissues and cells

2.1 Structure of mature tendon and ligament tissue—Tendon and ligament are dense, fibrous connective tissues that connect muscle and bone or bone and bone and transmit the mechanical forces that stabilize the skeleton and allow for body movement (10, 11). The ECM of the main part of the tendon and ligament (Figure 1A) consists predominantly of type I collagen (12, 13) and small amounts of other collagens, including

types III, IV, V and VI (14, 15). The proteoglycans found in tendon and ligament, include decorin, fibromodulin, biglycan and lumican. Their main function is to organize and lubricate collagen fiber bundles (14). Targeted disruption of these proteoglycans in mice leads to abnormal collagen fibrils in tendons (12, 13). The ECM of tendons and ligaments is hierarchically organized in several distinct layers (Figure 1A) (11, 16). Collagen microfibrils, the primary unit of mature tendon and ligament tissues, are first formed from collagen I molecules, organized as triple-helix polypeptide chains. Lateral and longitudinal stacking of microfibrils leads to the assembly of fibrils, which are arrayed in a parallel fashion to form collagen fibers with increasing diameter and mechanical strength. The fibers are then packed into larger units, termed fascicles, which are bundled by endotenon, and these bundles are wrapped by epitenon to form the complete tendon. Endotenon and epitenon are vascularized and innervated, which is essential for tendon homeostasis and response to injury (11, 16). The hierarchical structure of tendons and ligaments determines their biomechanical properties. The sharp ends of the fibrils, which on microscopy appear in wave-like or crimp patterns, provide tensile strength. The elasticity of tendons and ligaments is due to the large amount of type I collagen (17).

2.2 Biology of mature and progenitor tendon and ligament cells—The principal cell types in tendon and ligament are fibroblast-like cells, termed tenocytes, or ligament fibroblasts that are located between parallel chains of collagen fibrils (14, 21). Under normal conditions, tendon and ligament cells are thought to be quiescent and have very low rates of proliferation, while ECM production is responsive to changes in the mechanical load (22). These upregulated genes in tendon and ligament cells include types I, III, IV, V and VI collagen, which are also produced during embryogenesis to form tendon and ligament tissues. While these matrix genes are not specific to tendon and ligament cells, tenomodulin (Tnmd) is specifically expressed in mature tendon and ligament cells (23). However, its function in tendons and ligaments is not well characterized, and Tnmd knockout mice appear grossly normal (15).

The cell populations in tendon and ligament also include small subsets of progenitor cells (24). The deletion of biglycan and fibromodulin in mice causes the depletion of tendon progenitor cells, suggesting that these extracellular matrix components provide a niche environment for stem cells (12). The identification of specific molecular markers for tendon progenitor cells in adult tissues will help elucidate the contribution of this subset to tendon homeostasis and regeneration.

3. Developmental biology of tendon and ligament tissues and cells

3.1 Developmental biology of tendon and ligament tissues—Differentiation of tenocytes is tightly regulated by a series of growth factors and transcription factors (Figure 1B). The developmental origins of tenocytes and their interacting cells that initiate tenocyte differentiation differ in the trunk, limbs and craniofacial region (16). In the trunk, the tenocyte progenitor cells originate from the sclerotome, which is derived from somite, and they are condensed in a subregion called syndetome where *Scx* is expressed (25–27). The myotome, which is also derived from somite and the source of myocytes, is essential for the induction of tendon progenitor cells in the syndetome via FGF signals (27–29). In contrast,

tenocytes in limbs arise from the lateral plate mesoderm and interact with migrating myoblasts from the ventrolateral lip of the dermomyotome (25, 30). In this process, tenocyte progenitor cell induction does not depend on myocytes or myocyte-derived factors (31). At later stages, however, the absence of myoblasts causes disruption of tenocyte survival and tendon formation (25, 28, 32). In this process, TGF β and FGF signals in limb play a critical role in tenocyte differentiation by promoting Scx expression (28, 33). In addition, recent findings have revealed that the tendons in autopods differ significantly from those in zeugopods (34). While the development of the tendons in autopods depends on limb cartilage, that in zeugopods requires limb muscles to form properly.

An essential role of TGF β for tendon progenitor cell recruitment and maintenance during embryogenesis has been revealed by studies in mutant mice (28, 33). Tgfb2 and Tgfb3 double mutant mice or mice with TGFBR2 inactivation show the loss of tendons and ligaments in both the limbs and trunk (33). As TGF β and FGF are also involved in muscle and cartilage development (35), the specificity of these effects for tenogenic differentiation remains unclear; however, the focal and interactive effects of these multiple growth factor signals are integrated to promote proper tendon development (16). This notion is supported by the findings of a transcriptome analysis of limb tendons, which revealed that the expression of TGF β and FGF signaling related genes was highly upregulated during tendon differentiation and maturation (36).

Improvements in serial block face-scanning electron microscopy technology have shed new light on how the tendon anlage, which mainly consists of progenitor cells but less ECM, evolves into a mature tendon structure where tenocytes are embedded in collagen fibers. Kalson et al. stated that tendon development could be divided into two distinct stages: during the first stage (embryonic stage), tendon progenitors form cellular channels that define the extracellular space where collagen fibrils are deposited. This serves as the template for the second post-natal stage when the number of cells or collagen bundles remains constant and tissue growth depends on the increase in the diameter of collagen fibers (37).

3.2 Roles of transcription factors during tendon and ligament development—

Major advances in research on tendon and ligament biology include the recent identification of three transcription factors Scx, Mxk and Egr1 as critical regulators of tendon and ligament development and potentially homeostasis (Figure 1B). Scx is a helix-loop-helix (bHLH) transcription factor that is expressed in tendon progenitors (25, 26) and is required for the development of limb tendons. *Scx*^{-/-} mice show severe disruption of force-transmitting tendons (38). While the induction of tendon progenitor cells is normal in *Scx*^{-/-} mice, the formation of distinct tendons in the limbs, tail or trunk is disrupted. However, short tendons or ligaments that mainly act as anchors of bones or muscles are not affected by the loss of *Scx*, indicating an as-yet-undefined difference in the formation process of tendons and ligaments (38). The currently known targets of Scx-dependent transcription are *Col1a1*, *Col1a2*, *Acan* and *Tnmd* (23, 38–40). However, the *Scx* expression is strongly downregulated after birth (16), suggesting that Scx is essential for early stages of tendon development but may not be as important in the homeostasis of tendons and ligaments in adulthood.

The phenotypes of *Scx* and *Mkx* mutant mice are different. While *Scx*^{-/-} mutants exhibit loss of segments or of complete tendons (38), *Mkx*^{-/-} mice have reduced tendon mass without a decrease in the number of tendon cells (41–43). These data suggest that *Scx* is essential for the initiation of tendon differentiation, whereas *Mkx* plays a critical role in tendon maturation. Recently, *Mkx*^{-/-} rats have been generated using the CRISPR/Cas9 system (44). *Mkx*^{-/-} rats showed systemic hypoplasia of tendons similar to *Mkx*^{-/-} mice, and by virtue of their larger sizes, the physiological functions of tendon and ligament tissues were able to be evaluated in this model animal. A gait analysis revealed a significant decrease in the maximum ankle plantar flexion of *Mkx*^{-/-} rats (44). In addition, earlier heterotopic ossification of the Achilles tendon was observed in *Mkx*^{-/-} rats (44), which is discussed further in the **chondrogenic/osteogenic alteration of tendon and ligament tissues** section below.

Egr1 belongs to a family of zinc finger transcription factors named after the immediate response to growth factor stimulation (45). Among the four family members, *Egr1* and 2 are expressed in developing tendon tissue and play essential roles in tendon and ligament formation (46). Mice with mutations in either *Egr1* or *Egr2* showed a reduction of collagen fibrils numbers in embryonic tendons associated with a reduced expression of tenogenic genes, while misexpression of these genes led to the ectopic expression of *Scx* and *Col1a1* in a chick model (46). Interestingly, while the loss of *Egr1* resulted in a reduction in both *Scx* and *Mkx* expression in embryonic tendons, only the *Scx* expression and not the *Mkx* expression was impaired in adult *Egr1*^{-/-} injured tendon (47). These results indicate dynamic differences in the mutual regulation of these three transcription factors between embryonic and adult tendon and ligament tissue. Furthermore, no mice with knockout for these transcription factors showed complete loss of tendon or ligament tissue, implying that some functions may be shared among these factors or that as-yet-unknown factors ensuring the formation of tendon/ligament tissues may exist.

3.3 Transition zones between bone and tendon or ligament—The enthesis, which directly inserts into the metaphysis or diaphysis of long bones, is a heterogeneous composite of four zones (tendon and ligament ECM, fibrocartilage, mineralized fibrocartilage and bone), each having unique ECM and cell phenotypes (18, 19). This gradual change in structure is crucial to avoid rupture between two tissues upon application of a mechanical load. Signaling mechanisms that control the formation of the enthesis are being discovered. During the initiation stage of enthesis formation, *Scx*-dependent BMP4 expression in tenocytes is essential (49), and subsequent biomechanical stress caused by muscle contraction on this bone ridge area promotes further development of the tendon-bone connection (50). In addition, hedgehog (Hh) signaling is also indispensable for the proper maturation of enthesis tissue. Thomopoulos et al. found that the Hh signal is activated in a subpopulation of cells in the enthesis, and inactivation of Hh signaling in *Scx*-expressing cells impaired enthesis maturation (51).

Sox9 is a master regulator of chondrocyte differentiation and cartilage development. It starts appearing in mesenchymal condensations at the limb buds and induces chondrocyte progenitors by regulating cartilage-specific genes, such as *Col2a1*. In limbs, this chondrogenesis status is linked to tenocyte differentiation (52, 53). In the initiation phase,

Sox9 and Scx double-positive cells transiently appear in the mesenchyme and then transform to Sox9 or Scx single-positive cells (52, 53). The Sox9-Scx double-positive cells show delayed expression of the chondrocyte marker Col2, and molecular lineage tracing studies have shown that Sox9 and Scx double-positive cells mainly contribute to the tendon-bone ridge area, where Sox9 and Scx expression are specifically and exclusively restricted to chondrocytes and tenocytes at this stage. Loss of Sox9 (52, 53) and Scx (54) impaired the formation of the bone ridge and enthesis, indicating that both transcription factors are functionally essential for the formation of this transitional zone. This suggests that the balance between Sox9 and Scx regulates cell differentiation and the formation of the tendon-bone junction (52, 53).

Recently, Hh-activated and GDF5-positive cells were identified as enthesis progenitors (51, 55). Whether the classification of cell populations based on transcription factors (Sox9/Scx) and signaling molecules (GDF5/Hh) overlaps or represents different subsets is unclear at present. In this regard, GDF5-positive cells were found to be not a stable population but rather dynamically replaced by continuous recruitment and exit during joint formation (influx model) (56). Further investigation based on these findings will provide a more comprehensive understanding of enthesis development.

3.4 Transition zones between muscle and tendon—The transition zone between tendon and muscle is called the myotendinous junction (MTJ), a layer of ECMs including laminin, types I and IV collagen and thrombospondins. Cells in these areas interact with the ECM via transmembrane proteins, such as integrins and utrophin, that mediate intracellular signaling via dystrophin complex, talin/vinculin or integrin-linked kinase (20). Structurally, the MTJ forms a finger-like process between muscle sarcolemma and tendon collagen fibers to increase the attachment interface and provide resistance to the mechanical force. In mammalian MTJ, the ultrastructure of the finger-like process and its post-natal maturation is well described (57) (Figure 2), but the corresponding molecular and cellular mechanisms are less well-understood. Other model organisms, such as *Drosophila* and zebrafish, provide insight into this process. In zebrafish myosepta, secretion and accumulation of ECM between muscle segments occur first, followed by the recruitment of tenocytes and further maturation of ECM structure and binding with the muscle cells (58). Myosepta can be considered an analog of the mammalian MTJ. Most ECM components are common, and thus molecular mechanisms can also be conserved (58). For example, the functional importance of Col22a1 or thrombospondin4, ECM components found in mammalian MTJ, has also been proven in the zebrafish model system (59, 60). The craniofacial tendon of zebrafish can also serve as a model for mammalian long tendons (61). In addition to tissue formation, the mechanisms underlying the correct positioning and orientation of tendons and muscles, which derive from distinct progenitors, remain unclear in vertebrates (16). In *Drosophila*, an elaborate guiding system employing secreted Slit protein and Robo receptors governs muscle-tendon attachment (16). Slit ligand secreted from tendon cells is cleaved by the Robo2 receptor and oligomerized/stabilized on the tendon cell membrane, providing a repulsion/halt signal to muscle cells through Robo1/3 receptors (62).

3.5 Mechanobiology of tendon and ligament cells—Tenocytes respond to mechanical strain by increasing collagen production (22), and mechanical signals promote MSC differentiation into tenocytes (63). Mechanical stimulation also promotes collagen fiber thickening along with increased fiber density (64).

The three major tendon/ligament transcription factors (Scx, Mxk and Egr1) are also responsive to mechanical stimulation (65). Scx expression was promoted in bioartificial tendons composed of mouse mesenchymal stem cells after stretching *in vitro* and in the epitenon cells of mouse Achilles tendon after treadmill running *in vivo* (66). Mechanical force and misexpression of Scx synergistically promoted the differentiation of human MSCs to tenocytes (67). Egr1/2 was also increased after short treadmill exercise in rats (68), and the forced expression of Egr1 rescued tendon gene downregulation during tendon healing under conditions of a reduced mechanical load (69).

Mxk is also responsive to mechanical stimulation both *in vitro* and *in vivo*. In moderate treadmill exercise, the effects of mechanical loading on collagen fiber thickening are not observed in Mxk-deficient mice, suggesting that Mxk is critical in the response to mechanical stimuli (64). Interestingly, general transcription factor II-I repeat domain-containing protein 1 (Gtf2ird1) was found to play an important role in translating mechanical stress to Mxk expression (64). Gtf2ird1 is localized in the cytoplasm of unstressed tenocytes and translocated into the nucleus upon mechanical stretching to activate the Mxk expression, thereby acting as a mechano-sensor (64). GTF2IRD1 is 1 of 28 genes located in the Williams syndrome critical region (70), implying that this syndrome may be caused in part by dysfunction in this new mechano-signal cascade.

4. Pathogenesis of tendon and ligament tissues and cells

4.1 Aging and arthritis-associated changes in tendon and ligament tissues and cells—Histological abnormalities in ACL are highly prevalent in OA-affected knees and include cystic changes, the disorientation of collagen fibers and mucoid degeneration (9) (Figure 3). ACL degeneration can be initiated before or progress more rapidly than cartilage degeneration, at least in a subpopulation of individuals (71).

The earliest and most prevalent age-related change in the ACL extracellular matrix is the disorganization of collagen fibers, which can be seen in ACL from young donors without cartilage degeneration or inflammation (71). An aging-related decrease in the diameter of the collagen fibrils and a corresponding increase in the concentration of small fibrils have also been described for the human ACL (72).

The Mxk expression is maintained in murine and human mature tendon and ligament cells, and its expression is reduced in ACL cells from OA-affected joints (73). Indeed, Mxk knockdown in the ACL cells (annulus fibrosus cell and ligament-like cells in the inter-vertebral disc [IVD]), resulted in decreased tendon ECM gene expression (73, 74) (Figure 3). These findings suggest the potential role of Mxk in maintaining the tendon and ligament cell functions in adult tissues.

4.2 Chondrogenic/osteogenic alteration of tendon and ligament cells—The alteration of cell phenotypes is a main feature of injured or degraded tendons and ligaments (Figure 4). Chondroid metaplasia is a well-known feature of degenerated ACL (9). In this regard, *Mkx* mutant rats showed early heterotopic ossification in the Achilles tendon with elevated osteogenic and chondrogenic gene expressions (44). An analysis of tendon-derived cells revealed that *Mkx* deficiency accelerated chondrogenic and osteogenic differentiation, whereas *Mkx* overexpression suppressed chondrogenic, osteogenic and adipogenic differentiation (44) (Figure 4).

Under chondrogenic differentiation conditions, the stem/progenitor cells (TSPCs) from *Mkx* mutant rat tendon tissues (12) showed enhanced osteogenic and chondrogenic differentiation, and *Mkx* overexpression rescued this phenotype (44). These data suggest that *Mkx* has a dual function: maintain the production of tendon/ligament ECM components but also prevent the trans-differentiation of tendon/ligament cells to related lineage cells by inhibiting chondrogenic genes, including *Sox9*, and chondrogenesis. In support of this notion, a chromatin immunoprecipitation-seq analysis with rat primary tendon cells revealed that both tenogenic-related genes, such as *Col1a1* and *Col3a1*, and chondrogenic differentiation-related genes, such as *Sox5*, *Sox6* and *Sox9*, could be targets of *Mkx* (44).

4.3 Inflammatory signals in tendon and ligament cells—Biochemical mediators, such as cytokines and growth and differentiation factors, that are part of the abnormal synovial fluid composition in arthritic joints may also lead to alterations in the ligament cell phenotype. For example, IL-33 expression at the onset of tendinopathy leads to early tissue remodeling through type 1 to 3 collagen transition. Interestingly, IL-33 negatively regulates the expression of microRNA29a, which targets the IL-33 receptor *mST2* and *Col3a1* (75). Other studies have also addressed the role of inflammatory cytokines in tendon/ligament cell homeostasis and regeneration (76), mainly with regard to stimulating the production of inflammatory mediators and ECM-degrading enzymes.

IL-1 β is a main OA related inflammatory cytokine and inhibits the expression of *Mkx* as well as the expression of critical tendon/ligament-specific genes in cultured ligament cells. As described above, the expression of *Mkx* is reduced in human ACL from aged and OA-affected knees. One potential mechanism behind the suppression of *Mkx* is the involvement of inflammatory cytokines, such as IL-1 (73). Thus, the loss of *Mkx*, driven by proinflammatory cytokines such as IL-1 β , can lead to the abnormal chondrocyte-like differentiation of ligament cells and a reduced production of ECM with deficient biomechanical properties.

4.4 Pathogenic mechanical stress in tendon and ligament cells—Under ideal conditions, mechanical loading has a positive effect on gene expression and subsequent tendon and ligament tissue homeostasis. Tendon and ligament cells are highly sensitive to mechanical stimuli, and excess mechanical stimulation results in damaging the tendon and ligament tissues with a reduction in the tendon-related gene expressions. Indeed, extreme stretching and intense exercise have been reported to increase osteogenic and chondrogenic marker expression in tendons (66, 77).

While mild stretching of tenocytes *in vitro* increases the expression of Mx and tendon-associated genes, such as *Col1a1* and *Col3a1*, the same mechanical stretching stimulation of *Mx*^{-/-} tendon derived cells leads to chondrogenic differentiation with increased expression of chondrogenic markers, such as *Sox6*, *Sox9* and *Acan* (44).

The type of mechanical loading also affects the stiffness of tendons; stochastic strains are more likely to cause microdamage and decreased stiffness than cyclic strains (78). The mechanical loading factors, including amplitude, frequency, duration and period, are coordinately orchestrated for tendon homeostasis, and an imbalance in these mechanical signals leads to tendon and ligament tissue damage, partially via tendon and ligament cell catabolic gene and the Mx transcription factor network.

5. Tendon and ligament tissue engineering

Although tendon tissue has a regenerative capacity after injury at the neonatal stage, its potential to restore functional tissue is very limited in adults (79). Recent findings in tendon and ligament biology as summarized above have provided new opportunities to develop solid tissue engineering approaches to treating ligaments and tendons in adult (Figure 5). Bone marrow mesenchymal stem cell (MSC) transplantation can promote tendon injury repair (80, 81). To stabilize the transplanted MSCs and accelerate tissue regeneration, collagen scaffolds are successfully utilized both in tendon (80) and ligament (82) injury models.

To acquire more efficient regeneration of tendon and ligament tissues, pretreatment of MSCs with growth factors such as GDF5 and GDF7 (also known as BMP12) has been used to enhance the tenogenic differentiation of MSCs (80, 83–85).

The introduction of key transcription factors into MSCs is another promising strategy. MSCs with forced expression of Mx gain tendon and ligament cell shape and gene expression profiles and become resistant to the differentiation cues for the other cell types (74, 86, 87). When Mx-overexpressing MSCs were transplanted into injured IVDs *in vivo*, transplanted cells formed collagen fibrils with a similar diameter to the endogenous annulus fibrosus (AF) cells, the ligament-like cells in IVDs, and restored tissues were then protected against the mechanical stress to a similar degree as native tissue (74). These *in vitro* and *in vivo* findings indicate the potential utility of Mx in cell-based tendon and ligament tissue repair and engineering.

Egr1 also plays an important role in tendon and ligament differentiation (40) and in tendon gene expression (47). Furthermore, Egr1-expressing mesenchymal stem cells regenerated tendon tissue in a rat tendon injury model more efficiently than control mesenchymal stem cells (47). Scx, which had been considered to be more involved in the embryonic period than in later periods, also promoted more efficient tendon repair by TDSCs (88).

Mx, Scx and Egr1 may promote tendon/ligament cell differentiation of MSC not only by promoting tendon- and ligament-specific gene expression profiles but also by changing cellular responsiveness to other signaling cues. Egr1 may therefore contribute to Tgfb2 and Scx but not Mx expression (47). Mx-expressing MSCs showed increased responsiveness

for TGF β signaling and reduced responsiveness for BMP signaling, which promotes chondrogenic or osteogenic programs (74). The more optimized utilization of hierarchically or temporally controlled combinations of growth factors and transcription factors may substantially improve the tendon and ligament regeneration.

6. Conclusions and future directions

Recent advances in research on the biology of tendon and ligament cells, especially the identification and functional analyses of tissue-specific transcription factors, have shown novel mechanisms that control their differentiation and activation (Figure 5). However, important questions remain to be answered, such as how the hierarchical structure of tendon and ligament tissue is formed by tenocytes and how the precise and robust tissue integration occurs between tendon and ligament and bone or muscle. Emerging approaches such as mechano-transduction or knockout rat models should also provide new insight in the near future. While the myogenesis molecular program has been clarified by genome-wide analyses of DNA methylation with DNMT3A conditional knockout mice (89), further analyses of the tendon and ligament cell differentiation at the epigenetic level will be needed to establish a basis for tendon physiology and tissue engineering (89, 90). Progress in research on tendon and ligament biology will also shed light on its future therapeutic applications to musculoskeletal injuries and diseases. For example, correcting abnormal Mx expression may be a useful new approach to addressing tissue repair after injury and during chronic processes, such as OA. Pharmacological approaches to correcting abnormal differentiation through the inhibition of inflammatory signals or enhancement of differentiation cues can also be applied, such as using Mx expression as a parameter when screening for drug candidates. Further studies will be necessary, both in basic tendon and ligament biology as well as animal models of injury and disease, to achieve effective approaches for the treatment of tendon and ligament pathology.

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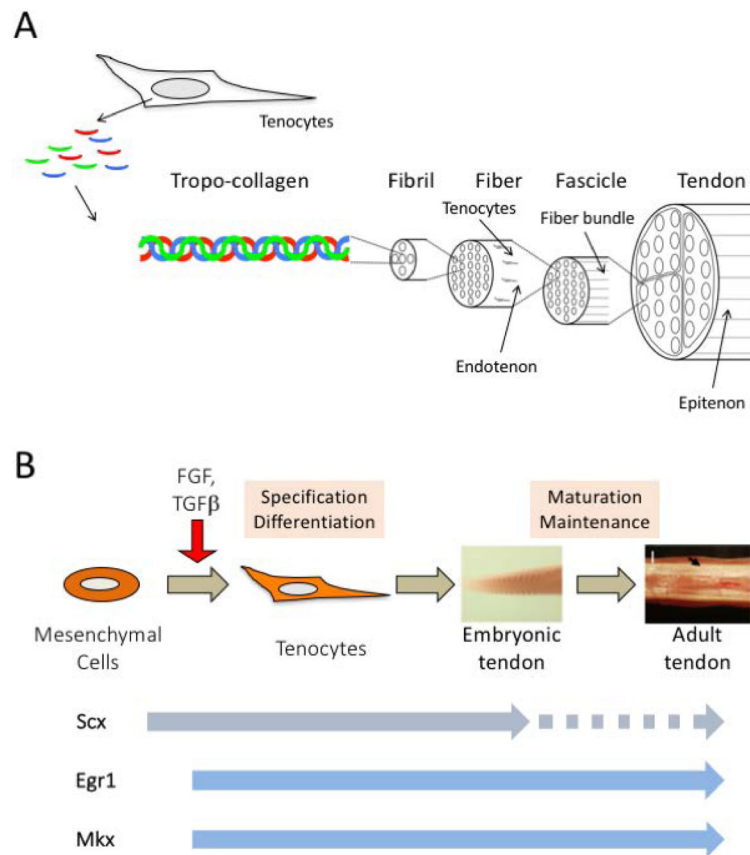


Figure 1.

(A) A schematic drawing showing the hierarchal structure of tendon/ligament tissue. Tropocollagens synthesized in tenocytes covalently bind and become collagen fibrils. The arrays of the collagen fibrils form collagen fibers. Tenocytes reside in the gaps of the fibers, connecting to one another through cellular channels. The collagen fibers are packed into fascicles that are bundled by endotenon. The fascicles are wrapped by epitenon to form the complete tendon/ligament tissue. (B) The expression period of the transcription factors and the signal input during the tendon development. Tenocytes are specified from mesenchymal progenitor cells by signaling cues, such as TGFβ or FGF, from neighboring tissues and marked by the expression of Scx. The expression of Egr1 and Mkx follow soon after and support the differentiation of tenocytes into embryonic tendon tissue. While the expression of Scx decreases after birth, the expression of Egr1 and Mkx persists throughout the post-natal maturation period and is essential for tissue maintenance during adult life. The panels are adopted from Itoh et al. PNAS 2010 and showing *in situ* hybridization of Mkx that indicates the embryonic tendon (left) and the tail tendon tissue from 10-week old mice (right).

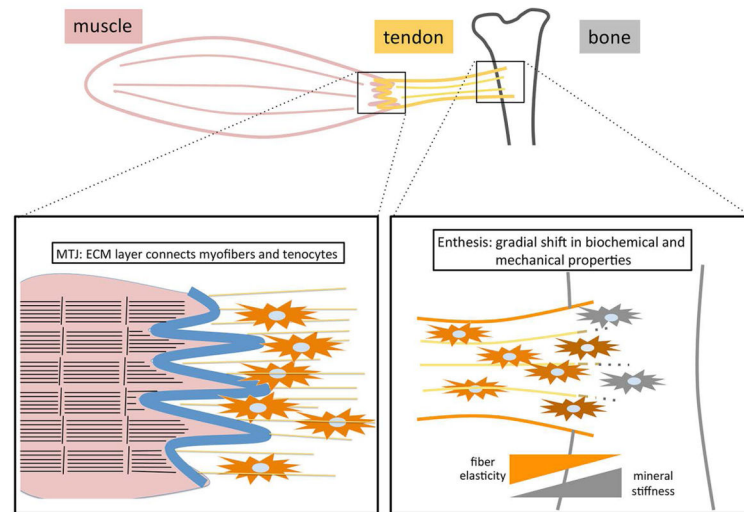


Figure 2.

The transition zones between tendon and muscle (MTJ) and tendon and bone (entheses). Top panel: a schematic drawing of the muscle-tendon-bone unit. Muscle, tendon and bone are depicted in red, yellow and gray, respectively. Bottom panel, left: MTJ. Muscle and tendon are illustrated on the left and right, respectively. Muscle sarcomeres are depicted as black horizontal lines, with vertical bars indicating Z lines. Collagen fibers are indicated as yellow lines, with tenocytes (orange cells) embedded. MTJ is the ECM layer with finger-like protrusions to increase the surface area between the muscle and tendon and is shown as a blue line. Bottom panel, right: entheses. Tendon and bone are illustrated on the left and right, respectively. The ECM contents and deposited minerals change gradually within the entheses, resulting in the continuous shift of tissue elasticity and stiffness.

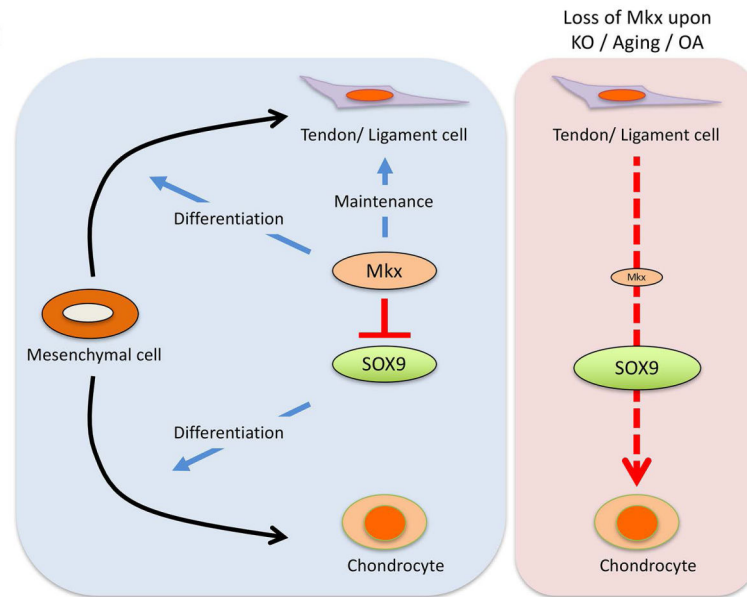


Figure 3.

Mxk is relevant for both the differentiation and the identity of tenocytes. Left: During embryonic development, mesenchymal progenitor cells acquire a tendon/ligament or chondrocyte cell fate, depending on lineage-specific transcription factors, such as Mxk or Sox9. In the adult tissue, the tendon/ligament cells continue to express Mxk, which is indispensable for the maintenance of their cellular identity, in part through the repression of Sox9 transcription. Right: When Mxk is genetically deleted or decreased by aging/OA, the SOX9 expression increases in tendon/ligament cells and these cells transform into chondrocyte-like cells.

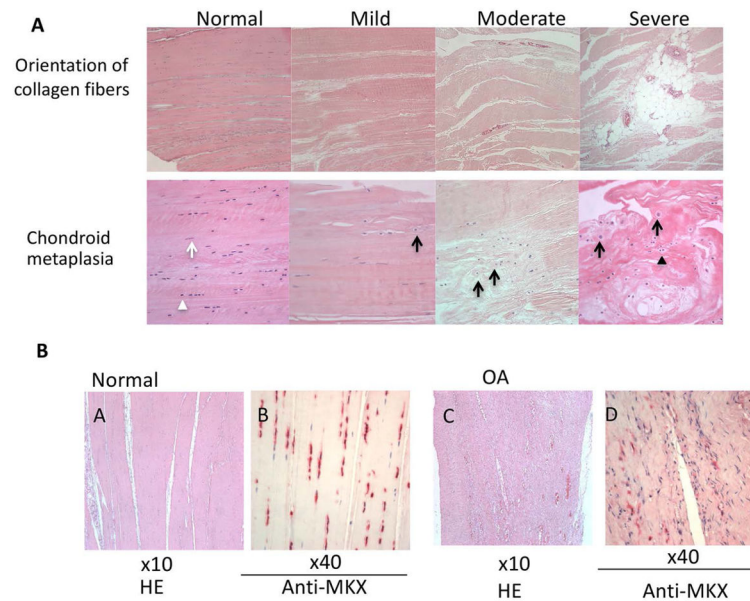


Figure 4.

A. Osteoarthritis (OA)-associated changes in the anterior cruciate ligament (ACL). Histological images of ACL from human knee joints illustrating changes in the collagen fiber orientation and chondroid metaplasia, the two most frequent changes. Chondroid metaplasia represents a shift in the ligament cell phenotype towards a more chondrocytic cell morphology. In normal ACL, only fusiform (white arrow) and ovoid cells (white arrow head) are observed. In mild cases, only a few spheroid cells are seen. In moderate cases, spheroid cells are dominantly observed. In severe cases, only spheroid cells (black arrow) and pseudo-cloning (black arrow head) that suggests chondroid metaplasia is observed and associated with mucoid degeneration. These histological changes in the ACL correlate with the severity of changes in the articular cartilage. B. MKX is expressed in normal ACL ligament cells. The numbers of ligament cells expressing MKX are reduced in OA ACL. HE staining (A, C) and MKX immunohistochemistry (B, D) (modified from Nakahara et. al. Arthritis Rheum 2013).

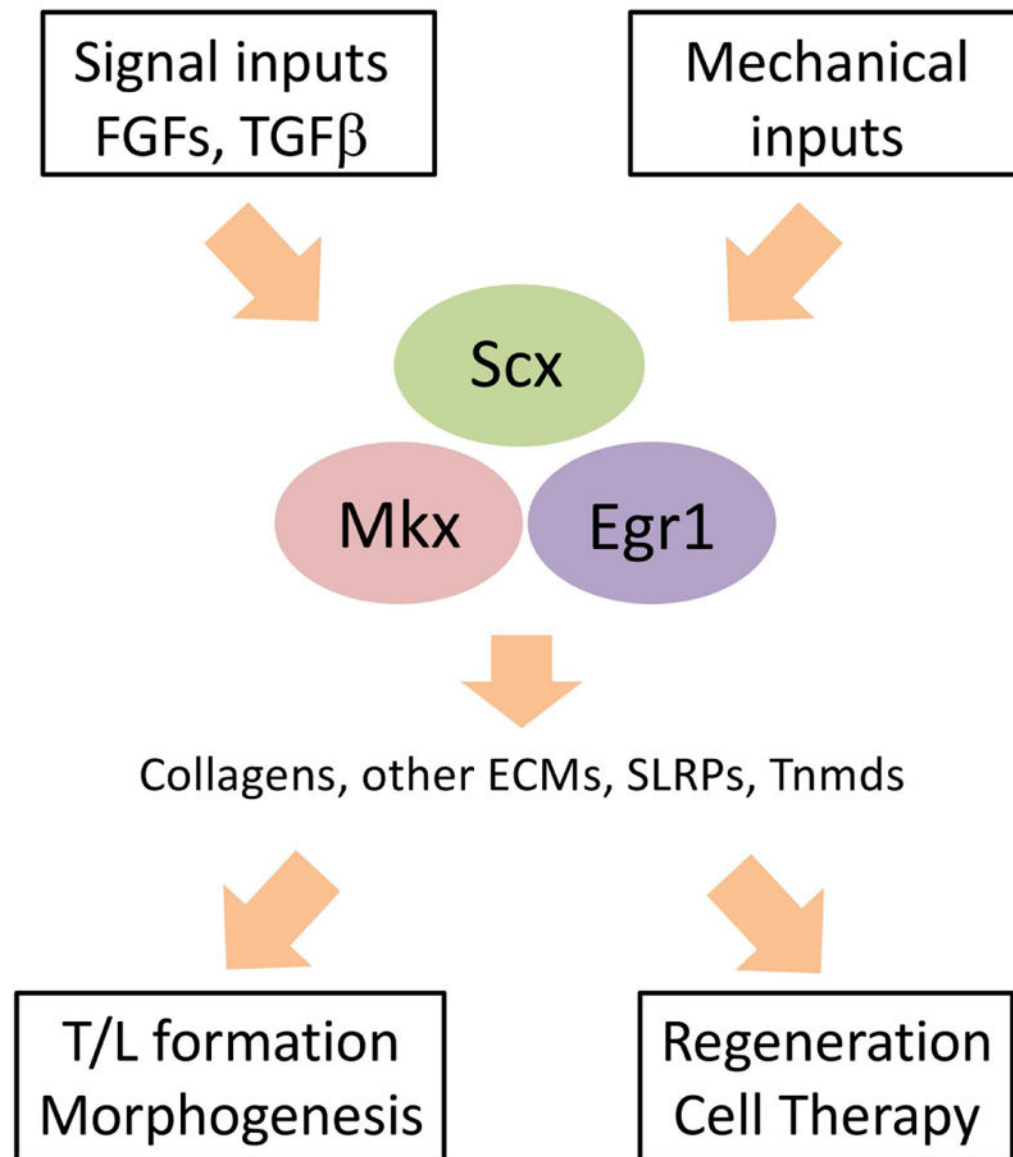


Figure 5.

A summary of the tendon/ligament biology discussed in this review. The expressions of Scx, Mkx and Egr1 are induced by signaling cues, such as TGFβ or FGFs (see also Figure 1), and promoted by mechanical cues. These transcription factors activate the transcription of tendon structural elements, such as collagens and SLRPs, which are indispensable for proper tendon and ligament tissue formation during embryonic development as well as tissue regeneration. They also play major roles in the maintenance of the tendon and ligament cell identity during aging or pathogenic processes (see also Figure 3), partly by repressing the factors, which promote other cell fates (see also Figure 4).